5.3.1 Gaussian elimination with row exchanges

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Week 05 due Feb 12, 2025 20:19 IST

Video

some other order, then you

don't change the solution to the linear system. So maybe it would it be possible

to reorder the equations in such a way that when you get to this particular

point, another element in a_21 below it, is actually on the diagonal. And if that

element is nonzero, then you be okay again. And the problem with that is that

you don't have the benefit of foresight, and you don't know how to reorder them

3:20 / 3:20

2.0x

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Reading assignment

1/1 point (graded)

Read Unit 5.3.1 of the notes. [LINK]

- [x] done

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Homework 5.3.1.1.

1/1 point (graded)

Perform Gaussian elimination as explained in Section 5.2.1 to solve

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

- [x] done

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Homework 5.3.1.2.

1/1 point (graded)

When performing Gaussian elimination as explained in Section 5.2.1 to solve

$$\begin{pmatrix} 10^{-k} & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

set the encountered computation

$$1 - 10^k$$

to

$$-10^k$$

(since we will assume k to be large and hence 1 is very small to relative to 10^k). With this modification (which simulates roundoff error that may be encountered when performance floating point computation), what is the answer?

- [x] $\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- [ ] $\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 - 10^{-k} \end{pmatrix}$
- [ ] $\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} \approx \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (when k is large)

Next, solve

$$\begin{pmatrix} 1 & 0 \\ 10^{-k} & 1 \end{pmatrix} \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

- [ ] $\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- [x] $\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 - 10^{-k} \end{pmatrix}$

$\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} \approx \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (when k is large)

Explain what you are observing.

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